

Seizures (epileptic seizures) are defined as transient symptomatic manifestations of abnormal excessive or synchronized neuronal activity of the brain.

They may be provoked due to acute cerebral insults (such as delirium, trauma or stroke), reflex (triggered by specific stimuli such as light or music), or unprovoked (epilepsy).

The International League Against Epilepsy (ILAE) has published a universal classification for seizures based primarily on consistent clinical observation of the seizure (semiology) and findings on ancillary tests such as the EEG and neuroimaging.

The ILAE classification (2017) distinguishes seizures into one of three types:

1. *Focal onset seizures* (previously: partial seizures) which originate within networks related to one hemisphere whether discrete or widely distributed within that hemisphere.
2. *Generalized onset seizures* (previously generalized seizures) which originate within and rapidly engage bilateral distributed networks.
3. *Unknown onset seizures* where the onset is not yet defined and a classification is not possible either due to a lack of observation of the onset or absence of a diagnostic evaluation (intended as a temporary category).

A 26-year-old woman reports brief episodes of déjà vu, rising abdominal feelings followed by confusion with lip smacking and right-hand fumbling. She has a history of febrile seizures as a child. Her brain MRI shows right mesial temporal sclerosis (MTS). A resting EEG shows sporadic right anterior temporal discharges and there is focal rhythmic activity originating over the right temporal channels during her episodes. Based on these focal features on seizure semiology (cognitive and autonomic aura at onset), neuroimaging (right MTS), and EEG (right temporal abnormalities), it may be inferred that her seizures originate within networks in the right hemisphere (mesial temporal lobe) – focal onset seizures.

On the other hand, A 17-year-old boy reports jerking and clumsiness in the morning, generalized convulsions precipitated by disco lights and binge

drinking. He has a family history of epilepsy. His brain MRI is normal and resting EEG shows generalized spike and wave discharges. An ictal EEG has not been obtained. Based on generalized features in his semiology (myoclonus and convulsions) and interictal EEG as well as the absence of a focal abnormality on neuroimaging, it may be inferred that his seizures either originate or rapidly engage within both hemispheres – generalized onset seizures.

Classification of seizures into focal or generalized onset is a crucial first step toward understanding the patient's epilepsy (disorder of recurrent seizures). The diagnosis of epilepsy is described in Chapter 20. Further classification of seizures is optional but may be done based on two important observations of semiology:

1. impairment in awareness at any time during the seizure
2. observation of movements or motor activity at the very onset of the seizure

The remainder of this chapter discusses the application of these observations [1].

Focal Seizures

Focal (onset) seizures are categorized based on impairment of awareness (anytime during event), motor manifestations (at the very onset), and bilateral progression.

Impaired Awareness

1. *Focal seizures without impaired awareness (focal aware seizures):* The patient is fully aware of themselves and their surroundings throughout the seizure even if immobilized. They were previously called “simple” partial seizures. Motor, sensory, autonomic, or psychic symptoms with preservation of awareness and consciousness (aura) are common forms. Some progress to impairment of awareness or spread bilaterally (secondary generalization). A majority of this seizure type do not have an associated surface EEG correlate (70%–90%) though focal abnormalities (interictal sharps or slowing) or ictal rhythmic activity may occur [2].
2. *Focal seizures with impaired awareness (focal impaired awareness seizure):* If the patient's awareness becomes impaired (not necessarily unconscious) at any point during the seizure it is called a focal seizure with impaired awareness. They were previously called “complex” partial seizures or dyscognitive seizures. Typically, they last about 30 s to 3 min and are accompanied by purposeless movements (automatisms), amnesia, and other features such as aphasia depending on the region of cortical involvement. Some progress to secondary generalization. The EEG almost always shows focal abnormalities (interictal sharps/slowing) or ictal rhythmic activity with bilateral spread [1,2].

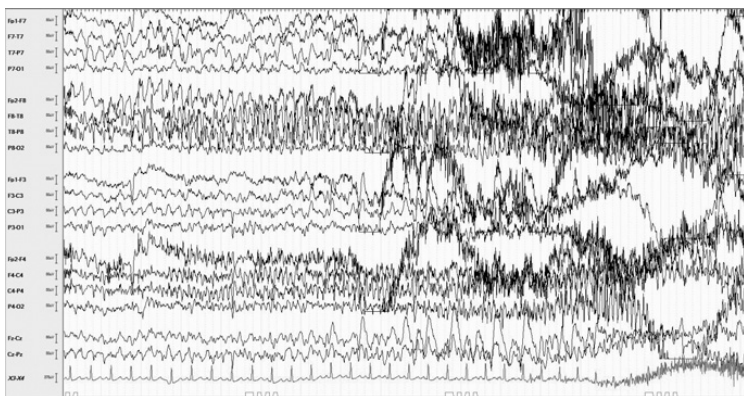


Figure 19.1 Twenty-three-year-old woman with temporal lobe epilepsy, focal onset (right hemisphere) evolves to bilateral tonic clonic (time base 20 s).

Motor Onset

1. *Focal seizures with motor onset (motor)*: If the seizure onset is characterized by motor movements (due to increase or decrease in muscle contraction) then it is called a motor onset seizure. Any motor activity after onset of the seizure does not qualify as a motor seizure. Common forms of motor activity include automatisms, tonic or clonic activity, myoclonus, hyperkinetic movements, atonic activity, and epileptic spasms. The EEG shows myogenic or movement artifact along with ictal activity.
2. *Focal seizures with nonmotor onset (nonmotor)*: If the seizure onset is not motor such as sensory, cognitive (aphasia), emotional, autonomic symptoms (aura), or behavioral arrest, then it is called a nonmotor seizure. The EEG may be normal or show ictal activity.

Focal to Bilateral Tonic Clonic

A focal onset seizure can spread bilaterally to become a bilateral tonic clonic seizure accompanied by loss of consciousness and convulsive activity (secondary generalization). The EEG shows spread of ictal activity into the contralateral hemisphere. Later parts of the record may be obscured by myogenic artifact [1]. Figure 19.1 shows bilateral tonic clonic evolution of a focal seizure.

Generalized Seizures

Generalized onset usually results in impaired awareness therefore these seizures are further classified based only on motor onset; not impairment of awareness.

1. *Generalized seizures with motor onset (motor)*: The seizure onset is characterized by generalized (often symmetric) motor activity:
 - *Tonic clonic (GTCs) seizures* involve loss of consciousness and tonic extension of limbs (usually less than a minute) followed by rhythmic clonic jerking of the extremities. There may be associated apnea, incontinence, and tongue biting followed by prolonged postictal confusion or stupor. EEG shows generalized ictal activity that may be completely obscured by myogenic artifact. Figure 19.2 shows the tonic phase of a GTC while Figure 19.3 shows its clonic phase.
 - *Tonic seizures* involve sudden stiffening and arm extension for a few seconds. They may lead to falls and injury. EEG shows brief attenuation and/or burst of high-frequency fast activity with myogenic artifact (tonic) as shown in Figure 19.4.
 - *Atonic seizures* involve abrupt momentary loss of tone associated with head drop, falls, and injuries. EEG shows brief attenuation and/or slow spike wave discharges without myogenic artifact (atonic) as shown in Figure 19.5.
 - *Myoclonic seizures* involve sudden brief (milliseconds) bilateral jerks (esp. arms) that occur in clusters that may be triggered by photic or sensory stimulation. EEG shows generalized polyspikes/wave discharges with intermittent myogenic artifact (myogenic).
 - *Myoclonic-atonic seizures* involve myoclonic seizures (often multiple) followed by an atonic seizure, that may result in a drop attack. EEG shows a generalized discharge (myoclonic component) followed by after going high-voltage slow wave (atonic component).

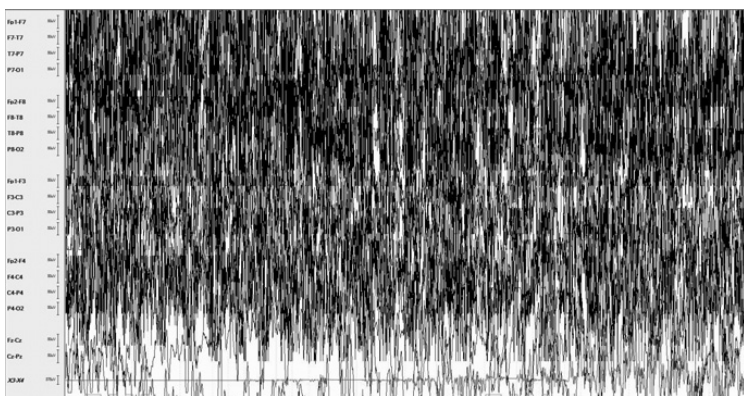


Figure 19.2 Same patient as in Figure 19.1, tonic phase of the GTC showing continuous myogenic artifact obscuring the recording.

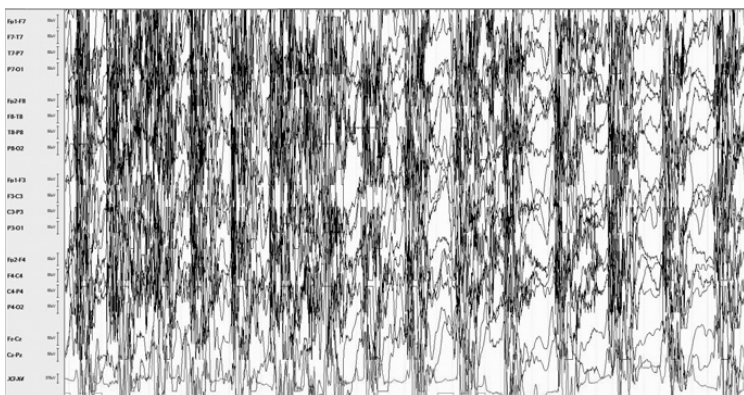


Figure 19.3 Same patient as in Figure 19.1, clonic phase of GTC showing periodic myogenic artifact obscuring the recording.

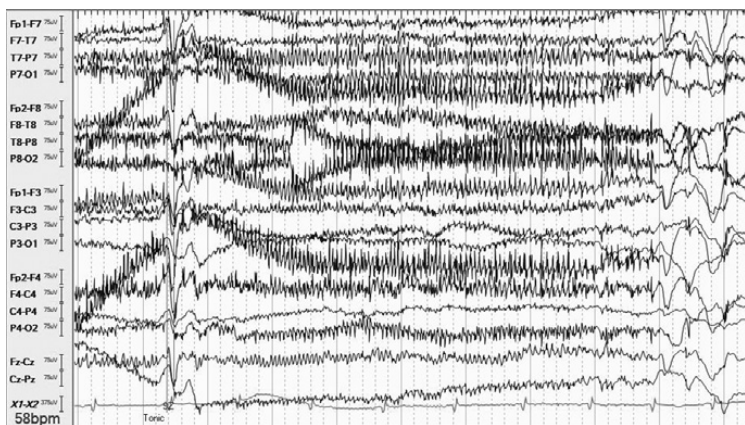


Figure 19.4 Fifty-four-year-old man with Lennox-Gastaut syndrome, tonic seizure characterized by brief generalized attenuation and low-voltage fast activity with overlying myogenic artifact.

- *Epileptic spasms* involve sudden brief (1–2 s) extensor, flexor, or mixed posturing of the proximal limbs or trunk in quick series. They may be subtle such as chin movements, head nodding, or grimacing. EEG abnormalities are enhanced with sleep or arousal and consist of high-voltage sharps/slow

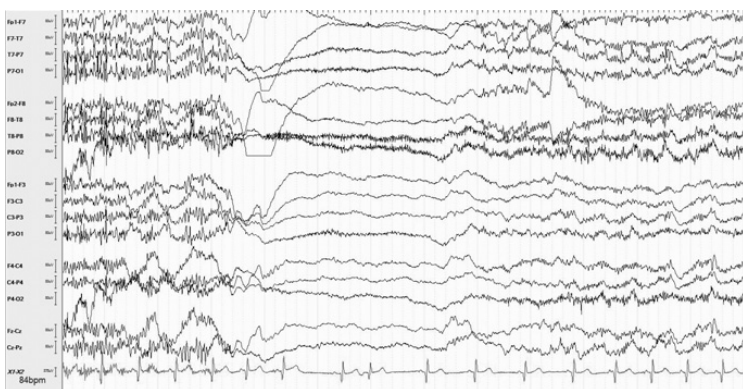


Figure 19.5 Same patient as in Figure 19.4, atonic seizure characterized by brief attenuation with slow spike and wave discharges. Note the absence of myogenic artifact.

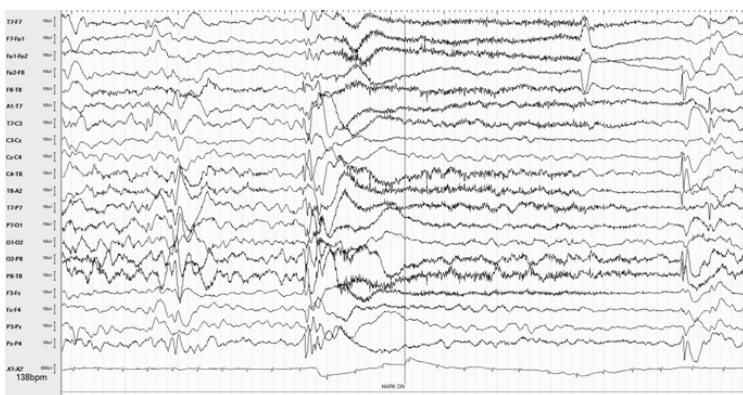


Figure 19.6 Nine-month-old boy with West syndrome, epileptic spasms characterized by high-voltage sharps/slow wave followed by low-amplitude fast activity and voltage attenuation (sensitivity of 20 μ V).

wave followed by low-amplitude fast activity and voltage attenuation as shown in Figure 19.6. It may be generalized (for generalized onset spasms) or focal (for focal onset spasms). These ictal patterns may occur multiple times in sleep without the accompanying clinical spasm.

2. *Generalized seizures with nonmotor onset (nonmotor)*: The seizure onset is characterized by the absence of motor movements, though some motor phenomena such as myoclonus or automatisms may follow. They were previously called absence seizures.

- *Typical absences* are characterized by abrupt onset and offset of impaired awareness usually associated with clonic movements of facial/perioral muscles. Absence status epilepticus may also occur. EEG shows regular 3 Hz generalized spike and wave discharges (GSW) in childhood absence epilepsy, faster irregular 3.5–6 Hz GSW and polyspikes and wave discharges may occur with later onsets. Figure 19.7 shows typical absences.
- *Atypical absences* are characterized by less abrupt onsets/offsets often associated with gradual head, truncal, or limb slumping and subtle myoclonus with only slight impairment in awareness. These are often seen in the intellectually impaired. EEG shows ongoing slow (less than 2.5 Hz) generalized spike and wave which is accentuated by hyperventilation. Figure 19.8 shows atypical absences.
- *Myoclonic absences* are characterized by rhythmic jerking of the arms, shoulders, and tonic abduction (arms lift up) with impaired awareness. EEG shows regular 3 Hz GSW with associated myoclonic artifact.
- *Absences with eyelid myoclonia* are characterized by brief, repetitive eye lid jerking with upward deviation of eyeballs and head extension. EEG shows high-amplitude generalized spike and wave/polyspikes and wave discharges at 3–6 Hz, provoked by photic stimulation. Figure 19.9 shows absences with eyelid myoclonia during photic stimulation.

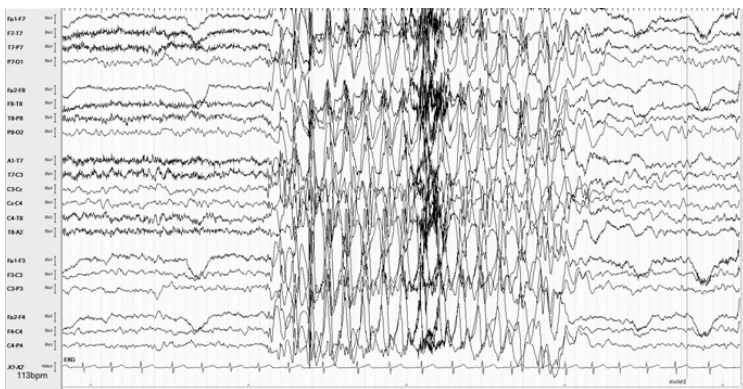


Figure 19.7 Ten-year-old girl with childhood absence epilepsy, typical absences showing a 3 Hz GSW discharges.

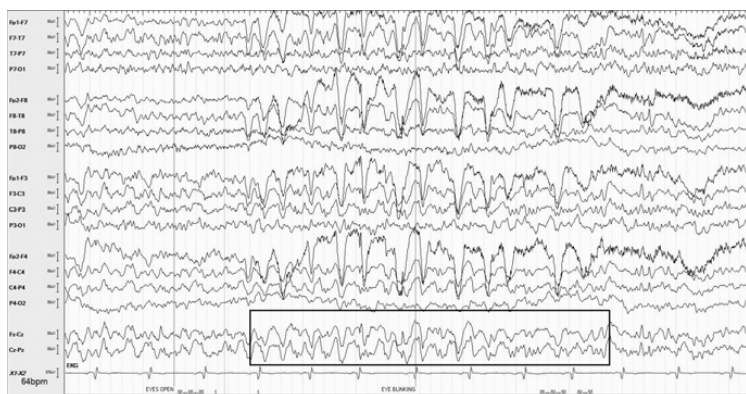


Figure 19.8 Same patient as in Figure 19.4, atypical absences with slow (2 Hz) generalized spike and wave discharges.

Neonatal Seizures

Seizures in neonates have focal onsets and are secondary to acute illness. They may be purely electrographic (particularly, in the critically ill) or associated with motor (such as automatisms, myoclonus, or spasms) or nonmotor onsets (autonomic or behavior arrest). For these reasons, neonatal seizures may not fit into the ILAE classification for older children and adults. The same clinical seizure type may at times, occur with or without electrographic correlate (electroclinical dissociation). Electrographically neonatal seizures are characterized by sudden, repetitive evolving stereotypical patterns with a definite onset and end but this definition has not been standardized. Their duration may be shorter than 10 s (such as BIRDs) [3]. Figure 19.10 shows a neonatal seizure.

The classification of seizure onset based on consistent clinical observation (focal or generalized onset), EEG and imaging findings forms the basis for diagnosing an epilepsy disorder as outlined in the following chapter (Chapter 20). Unknown onset seizures may also be classified as motor or nonmotor until their onset can be determined [1,4].

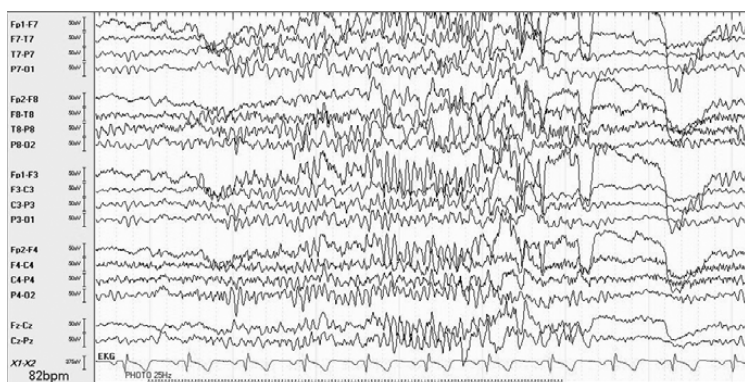


Figure 19.9 Eighteen-year-old woman with Jeavons syndrome, absences with eyelid myoclonia triggered by photic stimulation.

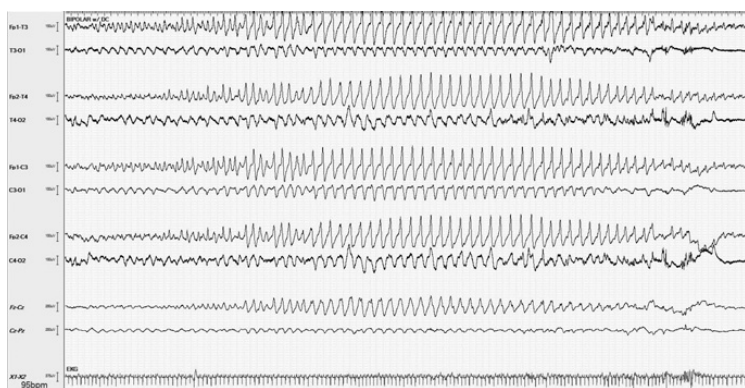


Figure 19.10 One-day-old boy with hypoxic ischemic encephalopathy, neonatal seizures characterized by evolving rhythmic activity (sensitivity 30 μ V, time base 2 min).

Chapter Summary

1. Seizures (epileptic) are manifestations of transient abnormal excessive or synchronized cerebral neuronal activity.
2. Seizures may be provoked (acute conditions) or unprovoked (epilepsy).
3. Seizures are classified as focal or generalized onset based on consistent clinical observation, EEG, and imaging findings.
4. Focal onset seizures originate from a single hemisphere, while generalized seizures originate from bilateral hemispheres.
5. Focal seizures may be further classified based on impairment of awareness (anytime during seizure) and motor or nonmotor activity (at the very onset).
6. Focal seizures without impaired awareness may not have surface EEG abnormalities.
7. Focal seizures may secondarily generalize; these are now called focal to bilateral tonic clonic seizures.
8. Generalized seizures are associated with impaired awareness, hence only motor or nonmotor activity at onset is used to classify them.
9. Common generalized motor seizures include generalized tonic clonic seizures (GTCs), tonic, atonic, myoclonic, myoclonic-atonic, and epileptic spasms.
10. Common generalized nonmotor seizures include typical and atypical absences, myoclonic absences, and absences with eyelid myoclonia.
11. Additional features of the seizure occurring after onset that may be relevant to its propagation can be added as descriptors.
12. If the onset of a seizure is yet undefined, it may be temporarily referred to as an unknown onset seizure until a further determination can be made about its probable onset.

References

1. Fisher RS, Cross JH, French JA, et al. Operational classification of seizure types by the International League against Epilepsy: position paper of the ILAE Commission for Classification and Terminology. *Epilepsia*. 2017 Apr;58(4):522–30.
2. Verma A, Radtke R. EEG of partial seizures. *Journal of Clinical Neurophysiology*. 2006 Aug 1;23(4):333–9.
3. Pressler RM, Cilio MR, Mizrahi EM, et al. The ILAE classification of seizures and the epilepsies: modification for seizures in the neonate. Proposal from the ILAE Task Force on Neonatal Seizures.
4. Berg AT, Berkovic SF, Brodie MJ, et al. Revised terminology and concepts for organization of seizures and epilepsies: report of the ILAE Commission on Classification and Terminology, 2005–2009. *Epilepsia*. 2010 Apr;51(4):676–85.